Type MS · Network protocols: what they are and which one does what

Answer 1 of 5 · Question 3 part c · 3 marks

2019 JUN · Paper 01 · extract 1 of 4 · find part c in this mark scheme extract

3(a)(i)	• LAN / Local area network (1) Allow PAN/Personal Area Network		1

Question Number	Answer	Additional Guidance	Mark
3(a)(ii)	• WAN / Wide area network (1)		1

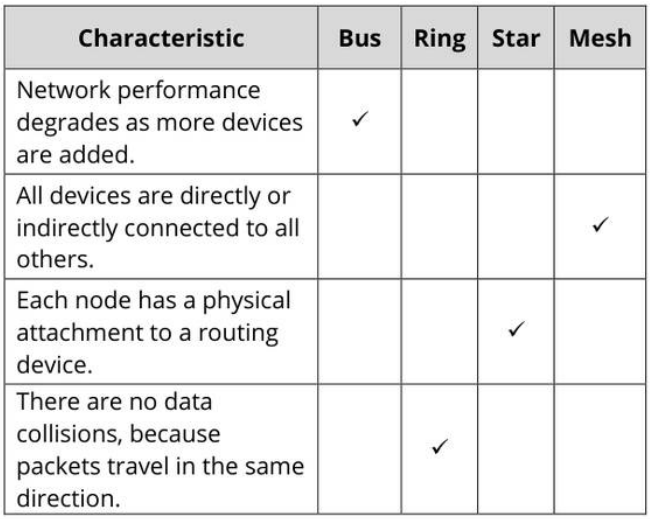
Question Number	Answer	Additional Guidance	Mark
3(b)	A: Megabits per second **The only correct answer is A** **B** is not correct because megapixels are associated with images rather than network data speeds **C** is not correct because mebicycles are not associated with network data speeds **D** is not correct because mebihertz are not associated with network data speeds		1

Type MS · Network protocols: what they are and which one does what

Answer 1 of 5 · Question 3 part c · 3 marks

2019 JUN · Paper 01 · extract 2 of 4 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(c)	• IMAP (1) AND **one** from: • because changes are synced (1) with the mail server in real-time (1) • because the messages always reside on the email server (1) as she has limited storage (1) on her tablet and phone • it gets sent to both smartphone and tablet (1) **Alternative answer:** • POP3 (not awarded a mark) AND **one** from: • because storage limitations on server (1) and privacy issues (1) • because more CPU time (1) may be used when syncing (1) (a large number of messages).		3

Question Number	Answer	Additional Guidance	Mark
3(d)			4

Characteristic	Bus	Ring	Star	Mesh
Network performance degrades as more devices are added.	✓			
All devices are directly or indirectly connected to all others.				✓
Each node has a physical attachment to a routing device.			✓	
There are no data collisions, because packets travel in the same direction.		✓		

Type MS · Network protocols: what they are and which one does what

Answer 1 of 5 · Question 3 part c · 3 marks

2019 JUN · Paper 01 · extract 3 of 4 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(e)(i)	-81 One mark for negative sign One mark for 81.		2

Question Number	Answer	Additional Guidance	Mark
3(e)(ii)	One mark for each nibble in correct order 0100 1011		2

Type MS · Network protocols: what they are and which one does what

Answer 1 of 5 · Question 3 part c · 3 marks

2019 JUN · Paper 01 · extract 4 of 4 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(e)(iii)	Any **two** from: 1. This is an overflow (1) error 2. It is caused by adding two 1s in the most significant bit (1) 3. The resulting number is too large to fit in the number of bits available / requires 9 bits / the number is larger than 2^8 / there is 1 that needs to be carried over (in MSB) / MSB gets discarded (1) 4. It can cause the program to crash or produce unreliable/incorrect results (1)		2

Question Number	Answer	Additional Guidance	Mark

Type MS · Network protocols: what they are and which one does what

Answer 2 of 5 · Question 1 part f.i · 2 marks

2020 NOV · Paper 01 · extract 1 of 5 · find part f.i in this mark scheme extract

1(a)	Any **two** from: • (Share) access to the Internet/WWW/broadband connection (1) • (Internal) communication (using email/instant messaging/calendars) (1) • Share files/data (1) • Share peripherals/printers / hardware (1) • Increases the amount of storage (since network storage devices can be attached) (1) • Saves money on licences (since network site licences for software are usually cheaper than buying a stand-alone licence for every machine) (1) • Centralised backup (1) • Centralised security (1)	• Award examples if mapped to a bullet point • Do not award the same bullet more than once • Do not award for install updates (in the stem)	2

Question Number	Answer	Additional Guidance	Mark
1(b) (i)	Any **two** from: • ~~Faster (data transfer) (1)~~ Not really true, its Greater bandwidth / more bits per second (1) • The connection does not get worse the further you are from the router / more reliable (1) • Connection does not get obstructed by walls, ceilings, and furniture (1) • More secure (1)	Accept reverse arguments for disadvantages of wireless.	2

Type MS · Network protocols: what they are and which one does what

Answer 2 of 5 · Question 1 part f.i · 2 marks

2020 NOV · Paper 01 · extract 2 of 5 · find part f.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(b) (ii)	Any **two** from: • Installation/maintenance is more complex (1) • Devices need to be physically connected (1) • Less portable / limited by length of cable (1) • Limited number of devices can be connected (1) • Some digital devices cannot use a wired connection (1) • Trip hazard (1)	Accept reverse arguments for advantages of wireless.	2

Question Number	Answer	Additional Guidance	Mark
1(c)	Award **two** marks for a linked description that addresses the individual mark points: The internet is a (global) network of networks/ connected devices (1) whereas the www is the collection of web pages/service accessed using the internet (1) / the www is resources located via URLs/domain names (1) The internet is the infrastructure (1) and the www is a service that runs on that infrastructure (1)		2

Question Number	Answer	Additional Guidance	Mark

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Type MS · Network protocols: what they are and which one does what

Answer 2 of 5 · Question 1 part f.i · 2 marks

2020 NOV · Paper 01 · extract 3 of 5 · find part f.i in this mark scheme extract

1(d)	Award **two** marks for a linked explanation such as: Running out of IPv4 addresses/the number of (internet) connected devices has grown (1) IPv6 addresses are long/longer than IPv4 / the number of possible addresses is large/will last much longer (1)		2

Question Number	Answer	Additional Guidance	Mark
1(e)(i)	Star		1

Question Number	Answer	Additional Guidance	Mark
1(e) (ii)	Any **one** from: • easy to connect/remove new nodes (1) • failure of one node/link does not affect the rest of the network (1) • easy to detect the failure of one node/link (1)		1

Question Number	Answer	Additional Guidance	Mark
1(e) (iii)	Any **one** from: • if central switch/hub fails then the whole network fails (1) • performance and number of nodes that can be added depends on capacity of central switch/hub (1) • can require more cable than some of the other topologies (1)		1

Question Number	Answer	Additional Guidance	Mark

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Type MS · Network protocols: what they are and which one does what

Answer 2 of 5 · Question 1 part f.i · 2 marks

2020 NOV · Paper 01 · extract 4 of 5 · find part f.i in this mark scheme extract

1(f)(i)	Any **two** from: • SMTP (1) • IMAP (1) • POP/POP3 (1)		2

Question Number	Answer	Additional Guidance	Mark
1(f)(ii)	Any **two** from: • passes the (reassembled) packets to the application layer (1) • check if all packets have arrived (1) • determine whether the contents are correct (1) • requests resending of lost or damaged packets (1) • reassembles packets in correct order/ into an email (1)	Do not accept receives from network layer (stem)	2

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Type MS · Network protocols: what they are and which one does what

Answer 2 of 5 · Question 1 part f.i · 2 marks

2020 NOV · Paper 01 · extract 5 of 5 · find part f.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Network protocols: what they are and which one does what

Answer 3 of 5 · Question 2 part b · 1 mark

2021 JUN · Paper 01 · extract 1 of 5 · find part b in this mark scheme extract

Number			
2(a)	Award **one** mark from: • share peripherals/devices (e.g. printers, scanners) (1) • communicate (e.g. email, instant messaging, play games) (1) • share data (e.g. files/music/videos/backups on servers) (1) • deploy/update applications/software(1) • administrative purposes (e.g. remote access/desktop)		1

Question Number	Answer	Additional Guidance	Mark
2(b)	Award **one** mark from: • to allow (connected) machines to communicate (1) • to provide the rules of communication between (networked) devices (1)	Accept any other response indicating protocols enable communication between devices	1

Type MS · Network protocols: what they are and which one does what

Answer 3 of 5 · Question 2 part b · 1 mark

2021 JUN · Paper 01 · extract 2 of 5 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	**The only correct answer is A** ***B*** *is not correct because FTP is an application protocol, which is added in the application layer* ***C*** *is not correct because FTP is an application protocol, which is added in the application layer* ***D*** *is not correct because FTP is an application protocol, which is added in the application layer*		1
2(c)(ii)	Award **one** mark from: • Adds the source/destination IP address (1) • Establishes sockets (an IP address followed by a port number) (1) • Moves packets onto the next network node (1) • Strips the source/destination IP addresses (when it arrives at its destination) (1) • Passes the packet to the transport/next layer (when it arrives at its destination) (1) • Adds/removes packet headers (1)		1

Question Number	Answer	Additional Guidance	Mark
2(d)	Award **one** mark for each of: **Domain Name** / **IP Address** / **URL** (blank) / (blank) / ✓ (blank) / ✓ / (blank) ✓ / (blank) / (blank)		3

Type MS · Network protocols: what they are and which one does what

Answer 3 of 5 · Question 2 part b · 1 mark

2021 JUN · Paper 01 · extract 3 of 5 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(e)(i)	Bus		1
2(e)(ii)	Award **one** mark from: • Easy to connect/remove devices/computers (in a network) (1) • Cable cost less / fewer cables needed (compared to mesh/star topology) (1) • Easier to plan/setup/maintain (1)		1
2(e)(iii)	Award **one** mark from: • If the main cable/a terminator fails or gets damaged the whole network will fail (1) • The more workstations the slower it gets (1) • The more workstations the more collisions (1) • Every device 'sees' all of the data on the network (1)		1

Question Number	Answer	Additional Guidance	Mark
2(f)(i)	Award **two** marks from: • Reduces time taken to upload/download (1) • Reduces the bandwidth required/cost of data to upload/download (1) • (Usually) produces smaller file/reduces storage (on the web server/device) (1) • People have a limited hearing range / can't tell the difference (1) • People's brains will 'fill in the blanks' (1)		2
2(f)(ii)	**The only correct answer is C** ***A*** *is not correct because it is not compressed* ***B*** *is not correct because it uses lossless compression* ***D*** *is not correct because it uses lossless compression*		1
2(f)(iii)	Award **one** mark from: • 3c3w2c6w3c (1) • c3w3c2w6c3 (1)		1

Type MS · Network protocols: what they are and which one does what

Answer 3 of 5 · Question 2 part b · 1 mark

2021 JUN · Paper 01 · extract 4 of 5 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(g)	Award **four** marks for a correct expression or correct answer **Examples of correct expressions:** 6 x 1000 x 1000 x 1000 x 8 (4) 54 x 1000 x 1000 1000 x 8 (4) 9 888.9 (4) For an incorrect or partial expression award **one** mark for each of: • **Bits to transfer** ○ 6×1000^3 (1) ○ x 8 (or equivalent) (1) • **Speed in bits per second** ○ 54×1000^2 (1) • Their bits divided by their speed (1)	• Units are not required • Equivalent expressions are accepted • Allow follow through for incorrect or partial expressions	4

Type MS · Network protocols: what they are and which one does what

Answer 3 of 5 · Question 2 part b · 1 mark

2021 JUN · Paper 01 · extract 5 of 5 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Network protocols: what they are and which one does what

Answer 4 of 5 · Question 4 part c.ii · 2 marks

2023 JUN · Paper 01 · extract 1 of 4 · find part c.ii in this mark scheme extract

4(a)	**The only correct answer is C** ***A*** *is not correct because* ***mebibits*** *is a measure of storage capacity* ***B*** *is not correct because* ***mebibytes*** *is a measure of storage capacity* ***D*** *is not correct because transmission speeds are measured only in bits per second, not* ***bytes per second***		1

Question Number	Answer	Additional Guidance	Mark
4(b)	Award **one** mark for each correct cell:		2

Type	Example	Number of bits
IPv4	192.169.0.3	32 (1)
IPv6	1050:a500:00c0:0440:0006:0300:700d:436f	128 (1)

Type MS · Network protocols: what they are and which one does what

Answer 4 of 5 · Question 4 part c.ii · 2 marks

2023 JUN · Paper 01 · extract 2 of 4 · find part c.ii in this mark scheme extract

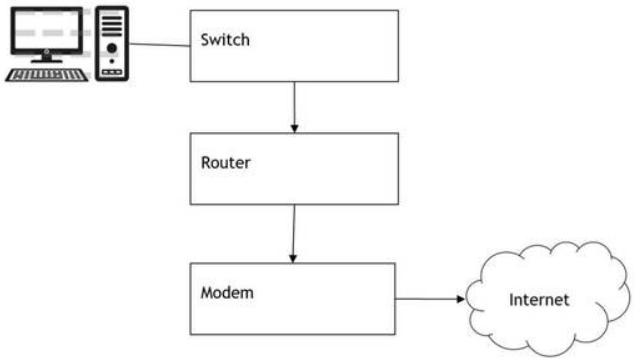
Question Number	Answer	Additional Guidance	Mark
4(c)(i)	Award up to **two** marks for a linked description, such as: • In a wired network packets travel over (copper/fibre-optic) cables (1), whereas in a wireless network packets travel over the air (1)	Allow radio signals, infra-red, microwave for air (must be more than just waves) Allow twisted pair, CAT for copper. Allow electrical/light signals (not just electricity)	2

Question Number	Answer	Additional Guidance	Mark
4(c)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. A protocol (suite/group/family) (1) 2. Used on a wired network / wired connection (1) 3. Defines physical parts/type of cable(twisted pair, CAT6)/type of connector (1) 4. Defines how packets are checked for errors (1) 5. Defines the speed of transmission (1) 6. Operates at the link layer of the TCP/IP stack (1)		2

Type MS · Network protocols: what they are and which one does what

Answer 4 of 5 · Question 4 part c.ii · 2 marks

2023 JUN · Paper 01 · extract 3 of 4 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(d)	Award **three** marks for all of switch, router and modem in the correct order as in diagram Award **two** marks for all of switch, router and modem in any order Award **one** mark for any of switch or router or modem in any box 		3

Type MS · Network protocols: what they are and which one does what

Answer 4 of 5 · Question 4 part c.ii · 2 marks

2023 JUN · Paper 01 · extract 4 of 4 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(e)	Award up to **two** marks for a linked description, such as: Validation is used to check that the data/password entered by a user is fit for processing/meets specified requirements (1), whereas authentication is used to verify the identity of the user/prove that users are who they say they are (1)	For validation 'specified requirements' a good description of the check can be given e.g. presence check, length check, format check	2

Question Number	Answer	Additional Guidance	Mark

Type MS · Network protocols: what they are and which one does what

Answer 5 of 5 · Question 1 part c · 1 mark

2025 JUN · Paper 01 · extract 1 of 4 · find part c in this mark scheme extract

<table>
<tr><td>1(a)</td><td>Award one mark for each correct row:

<table><tr><th>Statement</th><th>DNS</th><th>PAN</th><th>WAN</th><th>WWW</th></tr><tr><td>A network of networks not limited to a single location</td><td></td><td></td><td>✓</td><td></td></tr><tr><td>A network that connects devices within the range of a single user</td><td></td><td>✓</td><td></td><td></td></tr><tr><td>A set of hypertext documents accessed through the HTTP protocol</td><td></td><td></td><td></td><td>✓</td></tr><tr><td>A system that maps domain names to IP addresses</td><td>✓</td><td></td><td></td><td></td></tr></table></td><td>Do not award a mark for a row containing more than one tick.</td><td>4</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>1(b)</td><td>Award one mark for each of:

• Transport (layer) (1)
• (Data) link / network link / network interface (layer) (1)</td><td>Accept
• transfer or transmission for Transport

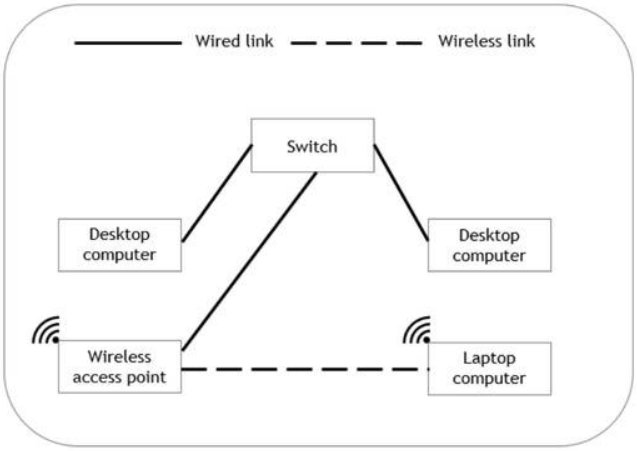
Do not accept:
• Network layer and Application layer (in the question)
• Data access layer, Internet layer (alternative names for network layer)
• Data’ or ‘network’ without ‘link’ or ‘interface’</td><td>2</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>1(c)</td><td>The only correct answer is D

A is not correct because HTTP is used to transfer hypertext
B is not correct because it is used to view email held on a server
C is not correct because it is used to download email from a server</td><td></td><td>1</td></tr>
</table>

Type MS · Network protocols: what they are and which one does what

Answer 5 of 5 · Question 1 part c · 1 mark

2025 JUN · Paper 01 · extract 2 of 4 · find part c in this mark scheme extract

Question	Answer	Additional Guidance	Mark
1(d)	Award **one** mark for each correct connection: • One solid line from wireless access point to switch (1) • One solid line from each desktop to switch (1) • One dashed line from laptop computer to wireless access point (1) 	No marks should be awarded if any additional lines have been drawn for each mark point. Do not double penalise (e.g. if you have penalised a second line from desktop to WAP then don't penalise extra line from WAP to desktop)	3

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Type MS · Network protocols: what they are and which one does what

Answer 5 of 5 · Question 1 part c · 1 mark

2025 JUN · Paper 01 · extract 3 of 4 · find part c in this mark scheme extract

Question	Answer	Additional Guidance	Mark
1(e)	Award up to **two** marks for a linked description such as: • To find weaknesses in a system (1) fix/repair weakness (1) • To find vulnerabilities in a system (1) by testing the effectiveness of the security measures in place (1) • An ethical hacking attack on a network (1) to identify vulnerabilities / recommend measures/take action to improve security (1) • Tests whether users of the network (1) can be tricked into giving away information / giving criminals access to the network / downloading malware (1) • To test the system security (1) / sanctioned by the owner (1)	Explanation can be a combination of different marking points. E.g. MP1a and MP2b	2
		Total for question 1	**12**

Type MS · Network protocols: what they are and which one does what

Answer 5 of 5 · Question 1 part c · 1 mark

2025 JUN · Paper 01 · extract 4 of 4 · find part c in this mark scheme extract

Question	Answer	Additional Guidance	Mark